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Chapter 03: Sequences of Real Numbers

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$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = e$$

$$\lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}$$



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Chapter 03 : Sequences of Real Numbers

I. Definitions and Notations

Definition

A numerical **sequence** is a function u defined by $\mathbb{N}$ or a subset of $\mathbb{N}$, in $\mathbb{R}$, which has the associated integer n , a real number $u(n)$, denoted u_n .

u_n the **general term** of the sequence.

A sequence of general term u_n is denoted by (u_n) .

Examples:

- a) $u_n = \frac{1}{n}, n \in \mathbb{N}^*$
- b) $u_n = 1 + (-1)^n, n \in \mathbb{N}$
- c) $u_n = \cos(n\pi), n \in \mathbb{N}$

Remark

A sequence can be defined by the value of the first terms and by a recurrence relation, that's to say, a relation linking several general terms of different ranks.

(u_n) is called a **recurrent sequence**.

Examples:

- a) $\begin{cases} u_0 = 1 \\ \forall n \in \mathbb{N}, u_{n+1} = 2u_n + 3 \end{cases}$
- b) $\begin{cases} u_1 = u_0 = 1 \\ u_n = u_{n-2} + u_{n-1} \end{cases}$ is called a **Fibonacci sequence**

Definitions

A sequence (u_n) is called

- 1) **Upper bounded**, if: $\exists M \in \mathbb{R}; \forall n \in \mathbb{N}, u_n \leq M$
- 2) **Lower bounded**, if: $\exists m \in \mathbb{R}; \forall n \in \mathbb{N}, u_n \geq m$
- 3) **Bounded**, if is upper bounded and lower bounded, or, $\exists K \in \mathbb{R}^*; \forall n \in \mathbb{N}, |u_n| \leq K$
- 4) **Increasing**, if: $\forall n \in \mathbb{N}, u_{n+1} - u_n \geq 0$ and **strictly increasing** if: $\forall n \in \mathbb{N}, u_{n+1} - u_n > 0$
- 5) **Decreasing**, if: $\forall n \in \mathbb{N}, u_{n+1} - u_n \leq 0$ and **strictly decreasing** if: $\forall n \in \mathbb{N}, u_{n+1} - u_n < 0$
- 6) **Monotone**, if : (u_n) is increasing or decreasing and **strictly monotone** if (u_n) is strictly increasing or is strictly decreasing.

II. Convergence

Definition

We say that a sequence (u_n) is **convergent** to the real number l if and only if:

$$\forall \varepsilon > 0, \exists n_0 \in \mathbb{N}; \forall n > n_0 \Rightarrow |u_n - l| < \varepsilon$$

We will then note $\lim_{n \rightarrow +\infty} u_n = l$.

Remark

A sequence that is not convergent is called **divergent**.

Examples:

- a) $\lim_{n \rightarrow +\infty} \frac{2n+1}{3n+2} = \frac{3}{2}$, $\lim_{n \rightarrow +\infty} (1 + e^{-n}) = 1$, $\lim_{n \rightarrow +\infty} (1 + e^n) = +\infty$
b) Show by definition that $\lim_{n \rightarrow +\infty} \frac{1}{n} = 0$, $\lim_{n \rightarrow +\infty} \frac{n+1}{n+2} = 1$

Proposition

- The limit of a sequence when it exists it is unique
- Any convergent sequence is bounded

Remark

A bounded sequence is not necessarily convergent, indeed, $u_n = (-1)^n$ is bounded and admits two limits **-1** and **1**.

Proposition

- An **increasing** and **upper bounded** sequence (u_n) is convergent
- A **decreasing** and **lower bounded** sequence (u_n) is convergent

Remarks

- 1) Any monotonic and bounded sequence is convergent
- 2) To study the nature of a sequence is to see whether it is convergent or not.

Arithmetic sequences and geometric sequences

Definition

- A sequence (u_n) is an **arithmetic** sequence if there exists a number r such that $u_{n+1} = u_n + r$ for all natural integers n . We then call r the **reason** of the sequence (u_n)

- Expression of the general term:** For $n \geq 0$; $u_n = u_0 + nr$.

- Expression of the sum of the first terms:** for $n \in \mathbb{N}$, we define S_n by $S_n = u_0 + u_1 + \dots + u_n$ then for all $n \in \mathbb{N}$, $S_n = \frac{n+1}{2}(u_0 + u_n)$ or

$$S_n = \frac{\text{Number of terms}}{2} (\text{first term} + \text{last term}).$$

- A sequence (u_n) is an **geometric** sequence if there exists a number q such that $u_{n+1} = qu_n$ for all natural integers n . We then call q the reason of the sequence (u_n)

- Expression of the general term:** For $n \geq 0$; $u_n = u_0 q^n$

- Expression of the sum of the first terms:** for $n \in \mathbb{N}$, we define S_n by

$$S_n = u_0 + u_1 + \dots + u_n \text{ then for all } n \in \mathbb{N}, S_n = u_0 \frac{1-q^{n+1}}{1-q}.$$

- Behavior to infinity:** Let (u_n) be a sequence of reason q and the first term u_0 then

- If, $q > 0$; $\lim_{n \rightarrow +\infty} u_n = \infty$

- If, $-1 < q < 1$; $\lim_{n \rightarrow +\infty} u_n = 0$

- If, $q < -1$; (u_n) has not limit

Example:

1) Let (u_n) be a sequence defined by

$$\begin{cases} u_0 = 1 \\ \forall n \in \mathbb{N}, u_{n+1} = \frac{1}{2}u_n + 3 \end{cases}$$

a) Calculate u_1, u_2, u_3

b) Prove by recurrence that $\forall n \in \mathbb{N}, u_n \leq 6$

c) Show that (u_n) is increasing

d) Deduce that (u_n) is convergent and calculate its limit

2) Let (v_n) be a sequence defined by $\forall n \in \mathbb{N}, v_n = u_n - 6$

a) Show that v_n is a geometric sequence

b) Calculate v_n then u_n as a function of n

c) Calculate $\lim_{n \rightarrow +\infty} v_n$ and $\lim_{n \rightarrow +\infty} u_n$

d) Calculate $S_n = v_0 + v_1 + \dots + v_n$ and $S'_n = u_0 + u_1 + \dots + u_n$

e) Calculate $\lim_{n \rightarrow +\infty} S_n$ and $\lim_{n \rightarrow +\infty} S'_n$

Subsequence

Definition

Let (u_n) be a sequence, we say that (v_n) is a subsequence or extracted sequence if there exists a function $K: \mathbb{N} \rightarrow \mathbb{N}$ strictly increasing such that $v_n = u_{K(n)}$.

Example:

$u_n = (-1)^n$, $u_{2n} = 1$ and $u_{2n+1} = -1$ are extracted sequences from (u_n)

Proposition

Every subsequence of a convergent sequence is convergent and has the same limit

Example:

The sequences $u_n = \frac{n+1}{2n+3}$, $u_{2n} = \frac{2n+1}{4n+3}$ and $u_{2n+1} = \frac{2n+2}{2n+5}$ have the same limit $\frac{1}{2}$.

Remark

A sequence that admits two subsequences that have two different limits is divergent

Example:

The sequence $u_n = (-1)^n + 4$ is divergent because $\lim_{n \rightarrow +\infty} u_{2n} = 5$ and $\lim_{n \rightarrow +\infty} u_{2n+1} = 3$

Theorem (Weierstrass-Bolzano)

Every bounded sequence admits a convergent subsequence.

III. Operations on convergent sequences

Definition

We define the sum, the product of two sequences (u_n) and (v_n) as well as the multiplication of a sequence (u_n) by a real number $\lambda \in \mathbb{R}$, by setting for all $n \in \mathbb{N}$:

- $(u_n) + (v_n) = (u_n + v_n)$
- $(u_n)(v_n) = (u_n v_n)$
- $\lambda(u_n) = (\lambda u_n)$
- $\left(\frac{1}{v_n}\right) = \frac{1}{(v_n)}, v_n \neq 0$

Theorem

Let (u_n) and (v_n) be two convergent sequences and $\lambda \in \mathbb{R}$, then

- $\lim_{n \rightarrow +\infty} (u_n + v_n) = \lim_{n \rightarrow +\infty} u_n + \lim_{n \rightarrow +\infty} v_n$
- $\lim_{n \rightarrow +\infty} (u_n v_n) = \lim_{n \rightarrow +\infty} u_n \lim_{n \rightarrow +\infty} v_n$
- $\lim_{n \rightarrow +\infty} \lambda u_n = \lambda \lim_{n \rightarrow +\infty} u_n$
- If $v_n \neq 0$ and $\lim_{n \rightarrow +\infty} v_n \neq 0$ we have $\lim_{n \rightarrow +\infty} \frac{1}{v_n} = \frac{1}{\lim_{n \rightarrow +\infty} v_n}$

Definition

Let (u_n) and (v_n) be two numerical sequences, we say that

- (u_n) equals (v_n) and we write $(u_n) = (v_n)$, if, and only if: $u_n = v_n, \forall n \in \mathbb{N}$
- (u_n) less than or equal to (v_n) and we write $(u_n) \leq (v_n)$, if, and only if:
 $u_n \leq v_n, \forall n \in \mathbb{N}$

Proposition

Let (u_n) and (v_n) be two convergent sequences, such that

$\exists n_0 \in \mathbb{N}^*; \forall n \geq n_0, u_n \leq v_n$ then $\lim_{n \rightarrow +\infty} u_n \leq \lim_{n \rightarrow +\infty} v_n$.

Proposition

Let (u_n) , (v_n) and (w_n) be three numerical sequences verifying

$\exists n_0 \in \mathbb{N}^*; \forall n \geq n_0, u_n \leq w_n \leq v_n$ then if (u_n) and (v_n) are convergent we have

$$\lim_{n \rightarrow +\infty} w_n = \lim_{n \rightarrow +\infty} u_n = \lim_{n \rightarrow +\infty} v_n.$$

Example: Calculate the limit of sequence (w_n) where

$$w_n = 2 + \frac{\cos(2n + 1)}{3n + 2}$$

Adjacent suites

Definition

Two numerical sequences (u_n) and (v_n) are said to be **adjacent** if one is **increasing** the other is **decreasing** and $\lim_{n \rightarrow +\infty} (u_n - v_n) = 0$.

Proposition

Two adjacent sequences are convergent and they have the same limits.

Example:

$$\left\{ \begin{array}{l} u_0 = 1 \\ u_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!} \end{array} \right. \text{ and } \left\{ \begin{array}{l} v_0 = 1 \\ v_n = u_n + \frac{1}{n!} \end{array} \right. \text{ are adjacent.}$$

IV. Cauchy sequence

Definition

A sequence (u_n) is Cauchy if, and only if,

$\forall \varepsilon > 0, \exists n_0 \in \mathbb{N}$; such that $\forall p, q \in \mathbb{N}$ checking $p > n_0$ and $q > n_0$ we have $|u_p - u_q| < \varepsilon$

That's to say $\lim_{p, q \rightarrow +\infty} |u_p - u_q| = 0$.

Remark

If $p > q$, then $\exists n \in \mathbb{N}^*$ such that $p = q + n$ so the last definition can be reformulated as follows:

(u_n) is a Cauchy sequence $\Leftrightarrow \forall \varepsilon > 0, \exists n_0 \in \mathbb{N}$; such that $\forall p \in \mathbb{N}$ checking $p > n_0, \exists n \in \mathbb{N}^*$ such that $|u_{p+n} - u_p| < \varepsilon$ That's to say $\lim_{p \rightarrow +\infty} |u_{p+n} - u_p| = 0$.

Proposition

Any Cauchy sequence is bounded.

Cauchy criterion

Theorem

A sequence is convergent if, and only if, it is Cauchy.

Example: $u_n = 2 + e^{-n}, v_n = 2 + e^n$.

Remarks

- a) Cauchy's criterion allows us to recognize that a sequence is convergent without needing to know its limit.
- b) We can know if the sequence does not converge by taking the logical negation.

(u_n) is **not a Cauchy sequence** $\Leftrightarrow \exists \varepsilon > 0$, such that $\forall n \in \mathbb{N}, \exists p, q \in \mathbb{N}, p > n, q > n$ and $|u_p - u_q| \geq \varepsilon$.

V. Upper limit and lower limit of a sequence

Definition

For a bounded sequence (u_n)

- a) We call the **upper limit** of (u_n) the limit of the sequence $v_n = \sup\{u_k, k \geq n\}$ and it is denoted $\lim_{n \rightarrow +\infty} \sup u_n$ or $\overline{\lim}_{n \rightarrow +\infty} u_n$.
- b) We call the **lower limit** of (u_n) the limit of the sequence $w_n = \inf\{u_k, k \geq n\}$ and it is denoted $\lim_{n \rightarrow +\infty} \inf u_n$ or $\underline{\lim}_{n \rightarrow +\infty} u_n$.

Examples:

$$u_n = (-1)^n, \quad \overline{\lim}_{n \rightarrow +\infty} u_n = 1 \text{ and } \underline{\lim}_{n \rightarrow +\infty} u_n = -1$$

$$u_n = 1 + 2 \cos n, \quad \overline{\lim}_{n \rightarrow +\infty} u_n = 3 \text{ and } \underline{\lim}_{n \rightarrow +\infty} u_n = -1$$

Remark

For a bounded sequence (u_n)

- a) $\overline{\lim}_{n \rightarrow +\infty} u_n$ and $\underline{\lim}_{n \rightarrow +\infty} u_n$ exist and they are unique, moreover $\underline{\lim}_{n \rightarrow +\infty} u_n \leq \overline{\lim}_{n \rightarrow +\infty} u_n$
- b) In general, $\overline{\lim}_{n \rightarrow +\infty} u_n \neq \sup u_n$ and $\underline{\lim}_{n \rightarrow +\infty} u_n \neq \inf u_n$.

Proposition

A bounded sequence (u_n) is convergent to l if, and only if $\overline{\lim}_{n \rightarrow +\infty} u_n = \underline{\lim}_{n \rightarrow +\infty} u_n = l$.

Theorem

Let (u_n) and $(\overline{u_n})$ be two bounded sequences and $\lambda \in \mathbb{R}$, then

- 1) If $u_n \leq \overline{u_n}$ then $\overline{\lim}_{n \rightarrow +\infty} u_n \leq \overline{\lim}_{n \rightarrow +\infty} \overline{u_n}$ and $\underline{\lim}_{n \rightarrow +\infty} u_n \leq \underline{\lim}_{n \rightarrow +\infty} \overline{u_n}$
- 2) $\overline{\lim}_{n \rightarrow +\infty} (u_n + \overline{u_n}) \leq \overline{\lim}_{n \rightarrow +\infty} u_n + \overline{\lim}_{n \rightarrow +\infty} \overline{u_n}$
- 3) $\underline{\lim}_{n \rightarrow +\infty} (u_n + \overline{u_n}) \geq \underline{\lim}_{n \rightarrow +\infty} u_n + \underline{\lim}_{n \rightarrow +\infty} \overline{u_n}$
- 4) If $\lambda > 0$; $\overline{\lim}_{n \rightarrow +\infty} (\lambda u_n) = \lambda \overline{\lim}_{n \rightarrow +\infty} u_n$ and $\underline{\lim}_{n \rightarrow +\infty} (\lambda u_n) = \lambda \underline{\lim}_{n \rightarrow +\infty} u_n$
- 5) If $\lambda < 0$; $\overline{\lim}_{n \rightarrow +\infty} (\lambda u_n) = \lambda \underline{\lim}_{n \rightarrow +\infty} u_n$ and $\underline{\lim}_{n \rightarrow +\infty} (\lambda u_n) = \lambda \overline{\lim}_{n \rightarrow +\infty} u_n$